

Erkan Alaydinov

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[X \(Twitter\)](#)

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Skills

- Deep knowledge of Solidity, EVM internals, smart contract architecture and common attack vectors.
- Daily use of Foundry for testing, fuzzing, symbolic execution and invariant/property-based testing.
- Strong understanding of DeFi protocols including lending, staking, AMMs, governance, yield aggregators, vaults, stablecoins, bridges and privacy protocols.
- Fluent in manual smart contract analysis, threat modeling, attack surface enumeration and exploit simulation.
- Strong understanding of AI-assisted smart contract security research.
- Extensive experience triaging and validating AI-generated security findings at scale, improving AI-driven security analysis through systematic validation and expert feedback.
- Experience fine-tuning AI models and building custom AI workflows for vulnerability research and audit automation.
- Strong understanding of DeFi mechanics: AMMs, oracles, leverage, liquidation, staking, bonding curves.
- Comfortable with Git-based workflows, diffing contract changes and version control under audit conditions.

Web3 Security Experience

Smart Contract Security Researcher | September 2023 - present

- Secured more than **70 Web3 protocols** across lending, staking, governance, AMMs, bridges, stablecoins, vaults, **yield aggregators**, privacy protocols and DeFi infrastructure.
- Audited **HyperLend (>\$1B TVL)** and multiple protocols exceeding **\$100M TVL**, including **Mystic Finance** and others.
- Led **41 private smart contract audits** as Lead Security Researcher.
- Audited over **100,000 nSLOC** across numerous production-ready smart contract systems.
- Performed large-scale **AI** security finding triage, reviewing and validating **3,000+ AI-generated findings** across dozens of protocols.
- Contributed to improving AI security models through high-quality vulnerability validation and structured feedback during model training.
- Worked closely with protocol teams throughout remediation, verification, and final security reviews.

Web3 Development Experience

Smart Contract Developer | castr.fun | 2026

- Helped design the **Castr.fun EVM** launchpad protocol and built core components of the end-to-end solution.
- Used Foundry to implement, test and debug
- Built major parts of the smart contract infrastructure.
- DeFi integrations with **Aerodrome** & blockchain transactions

Smart Contract Developer | Freelance / Various Protocols | June 2025 - Present

- Designed and implemented a complete **privacy-preserving donations protocol**, including smart contract architecture and core protocol logic.
- Contributed to the design and development of yield aggregation vault smart contracts, including vault architecture, strategy execution and protocol integrations.
- Used Foundry to implement, test and debug smart contracts.
- Developed smart contract integrations with **Aave** and **Morpho** protocols for yield aggregation strategies.
- Participated in architecture design, implementation, testing and optimization of production-ready smart contract systems.